LESSON 5.1

PLOTTING INTEGERS USING ONE-UNIT SCALES FOR AXES

LESSON INTRODUCTION

MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x- or y-axis as the line of reflection when two ordered pairs have an opposite x- and y-coordinate.

LEARNING TARGETS

- I can plot with integers using one-unit scales for axes.

BENCHMARK COHERENCE

BUILDING ON

MA.5.GR.4.1 Identify the origin and axes in the coordinate system. Plot and label ordered pairs in the first quadrant of the coordinate plane.

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How can you use ordered pairs to locate points in the coordinate plane?
- How do coordinate grids organize information?

LESSON NARRATIVE

Students investigate the relationship between a point's location in the coordinate plane and the structure of its coordinates. Students then draw conclusions about the signs of the coordinates for ordered pairs in different quadrants. Then students explore how a reflection over the x- or y-axis is performed and its effects on the coordinates of a point. This lesson is accompanied by a Desmos Activity Builder.

COMPONENT SUMMARY

5.1.1 WARM-UP *(~5 MIN.)*

Students predict the number of pieces of candy corn shown in a picture

5.1.2 EXPLORATION *(10-15 MIN.)*

Desmos Activity Builder connecting ordered pairs to their location on the coordinate plane

5.1.3 BRING IT TOGETHER *(5-10 MIN.)*

Summary activity on how to read, locate, and write ordered pairs

5.1.4 GUIDED INSTRUCTION *(10-15 MIN.)*

Students plot ordered pairs and investigate the coordinates of reflections on the coordinate plane

5.1.5 GUIDED PRACTICE *(5 MIN.)*

Practice identifying the coordinates of reflections of coordinate points

5.1.6 WRAP-UP *(~5 MIN.)*

Formative assessment on identifying the coordinates of ordered pairs and their reflections

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Coordinate plane
- Quadrant
- Ordered pair

MATERIALS

- (Optional)Tape for axes
- (Optional) Highlighters

ADDITIONAL PREPARATION

- 5.1.1 Warm-Up includes a video to accompany the questions.
- Desmos Activity Builder is available to accompany this lesson.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may plot ordered pairs (y,x) instead of (x,y).
- Students may have trouble distinguishing reflecting over the x-axis versus the y-axis.

THINGS TO CONSIDER

- The Desmos Activity provides an opportunity for students to digitally interact with the content and each other. The activity builder could be utilized in a whole group setting, for students to explore in small groups, or a combination of both.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

5.1.2 Exploration positions students to notice the coordinates of points that lie on the x- and y-axes. Consider applying Notice and Wonder routines on each set of images, positioning students to discover the relationship between a point's coordinates and its location in the coordination plane.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

In the 5.1.2 Exploration, students will engage in mathematical discussion with their peers about the coordinate grid and placement of ordered pairs. Students work with their partner to summarize the characteristics of points in each quadrant in writing. Clarifications within this standard emphasize developing students' abilities to justify their thinking and compare their responses to responses of peers.

DIFFERENTIATE INSTRUCTION

Prompt students to write out the coordinates of a point they must consider on the coordinate plane. By adding this step to an activity, students have an opportunity to self-check that they have plotted a point correctly. Furthermore, it helps them to solidify connections between a point's location, coordinates, and quadrant. 5.1.2 Exploration implements visual entry points into discussion about the coordinate plane. Consider using the interactive coordinate plane features in the Desmos lesson to provide additional visuals.

5.1.1 WARM-UP: GUESS THE NUMBER

Component Overview: *Display the image below for the entire class. After students discuss their estimates, play the video showing the accurate number of candy corn pieces. You can also show the still image of the video for a few moments before playing for students to estimate.*

TEACHER MOVES

MA.K12.MTR.4.1: Discussion and Reflection

After students make an estimate independently, place students in groups of three (if possible) to discuss their strategies for estimating. Consider structuring the discussion so each student has a set amount of time to explain their thinking and for their partners to summarize the justification. When sharing whole group, consider asking students to share a justification from a peer that they found interesting.

5.1.1 Warm-Up: Guess the Number

A picture of $\frac{1}{4}$ cup of candy corn is shown.

1. Estimate the number of pieces of candy corn that are in the measuring cup.
   Answers will vary. I think the cup holds 12 or 20 pieces of candy corn.

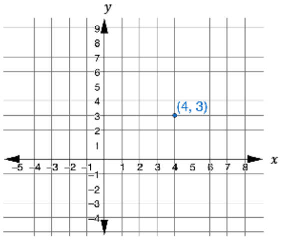
2. Ask two classmates the value they estimated. Record their value and write your summary of their reason. You are the only person who should write in the first two columns of the table. Have the person initial next to your summary stating that your summary is correct.
   Answers will vary.

Guess	My Summary of Their Reason	Initials
12	*Answers will vary. I tried counting what I saw in the picture.*	*CL*
15	*Answers will vary. I multiplied what I think each row holds by each column.*	*SS*

3. How close was your estimate to the actual number?
   Answers may vary. My estimate was higher by 4 candy corn because I thought the dish held more space.

5.1.2 Exploration: The Coordinate Plane

1. This graph shows a point that is plotted on a coordinate plane.



The two numbers are called the coordinates. The two numbers are also known as an ordered pair, (x, y).

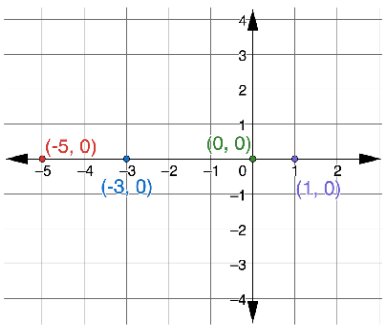
   a. Discuss with your partner which coordinate will change if the point is dragged from left to right.
   The x-coordinate will change as the point is dragged from left to right. On a horizontal number line the numbers increase as you move to the right and decrease as you move to the left.

   b. Discuss with your partner which coordinate will change if the point is dragged up or down.
   The y-coordinate will change as the point is dragged up and down. On a vertical number line, the numbers increase as you move up and decrease as you move down.

   c. If the point $(4, 3)$ could be dragged around on the coordinate grid, what changes would you expect to see in the ordered pair? Select all that apply.

   - ☐ The x-coordinate becomes smaller when the point is moved to the left.
   - ☐ The x-coordinate becomes larger when the point is moved to the left.
   - ☐ The x-coordinate stays the same when the point is moved to the left.
   - ☐ The y-coordinate becomes smaller when the point is moved up.
   - ☐ The y-coordinate becomes smaller when the point is moved down.
   - ☐ The y-coordinate stays the same when the point is moved up.

2. Four points are plotted on the coordinate grid.



   a. What do you notice about all four points?
   Answers will vary. All of the four points have the same y-coordinate of 0 and are on the x-axis.

5.1.2 EXPLORATION: THE COORDINATE PLANE

Component Overview: *Students will be exploring the relationship between an ordered pair's location and the structure of its coordinates. The first exploration helps students develop the understanding that the x-coordinate represents a point's horizontal position, and the y-coordinate represents a point's vertical position. Then, students investigate the coordinates of ordered pairs located on the axes. Finally, students make a connection between the signs of the coordinates and which quadrants they lie in.*

TEACHER MOVES

Consider providing sentence stems to help structure partner discussion.

"As the point moves from left to right, the _____ coordinate will __________."

"As the point moves up and down, the _____ coordinate will __________."

TEACHER MOVES

Allow students time to discuss incorrect answers with a partner. Also prompt students to think of edits they could make to some answer choices (like the second and fourth choice) that would make them correct.

DIFFERENTIATE INSTRUCTION

Consider presenting the animation found in the Desmos Activity for question 2. Also consider turning the classroom into a coordinate plane using tape to create the axes. Allow students to walk along the axes and throughout the quadrants to help build understanding.

5.1.2 EXPLORATION: THE COORDINATE PLANE (CONTINUED)

TEACHER MOVES

Notice and Wonder

Engage students in a Notice and Wonder routine to position them to draw conclusions about plotting points in the coordinate plane.

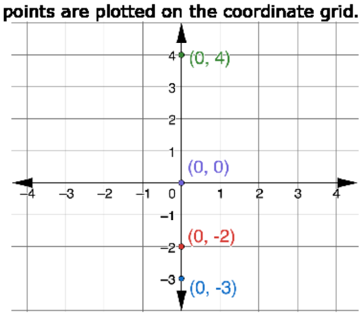
- What do you notice?
- What is the same? What is different?
- What does that make you wonder or think about the coordinates of a point either on the x-axis or on the y-axis?

b. What is the same about the points?
Answers will vary. All of the four points have the same y-coordinate of 0 and are on the x-axis.

c. What is different about the points?
All four points have different x-coordinates.

d. Discuss with your partner what is true about all the points on the x-axis.
All points on the x-axis have the same y-coordinate, and the y-coordinate will be 0.

3. Four points are plotted on the coordinate grid.



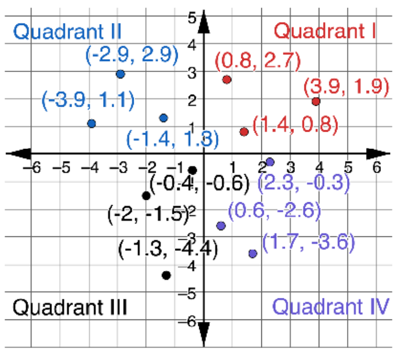
a. What is the same about the points?
Answers will vary. All four points are on the same line, the y-axis, and have the same x-coordinate.

b. What is different about the points?
Answers will vary. Each point has a different y-coordinate.

DIFFERENTIATE INSTRUCTION

Consider allowing students alternate ways to record their summaries in the table aside from a written explanation. For example, provide a template like this: (_______,_______) so students can record the signs of the coordinates in the appropriate spots.

4. Twelve points are plotted on the coordinate grid. The grid shows the labels Quadrant I, Quadrant II, Quadrant III, and Quadrant IV.

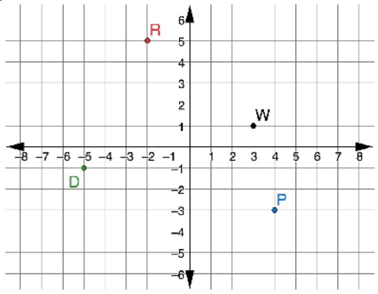


Work with your partner to summarize the characteristics of the points in each quadrant. Write your summary in the table.
Answers will vary.

Quadrant I	*The x- and y-coordinates in this quadrant are both positive.*
Quadrant II	*The x-coordinates in this quadrant are negative. The y-coordinates are positive.*
Quadrant III	*The x- and y-coordinates in this quadrant are both negative.*
Quadrant IV	*The x-coordinates in this quadrant are positive. The y-coordinates are negative.*

5.1.3 Bring it Together: Plotting Ordered Pairs

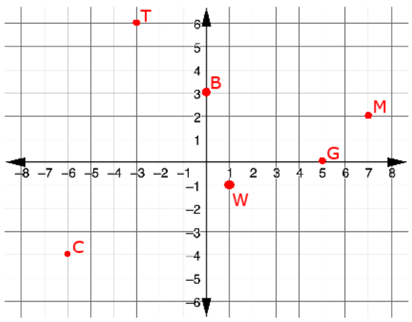
1. Points D, P, R, and W are plotted on the coordinate grid.



Select all of the descriptions of each point.

Point D	Point P
☐The x-coordinate of D is -5.	☐The x-coordinate of P is -3.
☐The x-coordinate of D is -1.	☐The x-coordinate of P is 4.
☐The y-coordinate of D is -5.	☐The y-coordinate of P is -3.
☐The y-coordinate of D is -1.	☐The y-coordinate of P is 4.
☐Point D is in quadrant I.	☐Point P is in quadrant I.
☐Point D is in quadrant II.	☐Point P is in quadrant II.
☐Point D is in quadrant III.	☐Point P is in quadrant III.
☐Point D is in quadrant IV.	☐Point P is in quadrant IV.
Point R	**Point W**
☐The x-coordinate of R is -2.	☐The x-coordinate of W is 1.
☐The x-coordinate of R is 5.	☐The x-coordinate of W is 3.
☐The y-coordinate of R is -2	☐The y-coordinate of W is 1.
☐The y-coordinate of R is 5.	☐The y-coordinate of W is 3.
☐Point R is in quadrant I.	☐Point W is in quadrant I.
☐Point R is in quadrant II.	☐Point W is in quadrant II.
☐Point R is in quadrant III.	☐Point W is in quadrant III.
☐Point R is in quadrant IV.	☐Point W is in quadrant IV.

2. Use the coordinate grid to plot and label points that have the characteristics described in the table.



Answers will vary.

Point B has an x-coordinate of 0. *Example shown on graph above.*	Point C has a y-coordinate of -4 and is in Quadrant III. *Example shown on graph above.*
Point G has a y-coordinate of 0. *Example shown on graph above.*	Point M has a y-coordinate of 2 and is in Quadrant I. *Example shown on graph above.*
Point T has an x-coordinate of -3 and is in Quadrant II. *Example shown on graph above.*	Point W has an x-coordinate of 1 and is in Quadrant IV. *Example shown on graph above.*

5.1.3 BRING IT TOGETHER: PLOTTING ORDERED PAIRS

Component Overview: *Students identify the x- and y-coordinates of ordered pairs and their corresponding quadrants. Students then plot ordered pairs after being given instructions about their coordinates or location.*

TEACHER MOVES

Encourage students to share out explanations for the selections in this activity. Use this as an opportunity to clarify misconceptions about whether the x- or y-coordinate comes first in an ordered pair.

DIFFERENTIATE INSTRUCTION

Provide additional auditory entry points for this section to ensure all students have access to the content. It may be beneficial to have students work in pairs to read the descriptions aloud or to pull a small group of struggling readers to work with while the rest of the class works independently.

ENRICHMENT

If you notice students struggling to plot the points, ask questions to provide further scaffolding:

- Where does a point fall if it has an x-coordinate of zero? Why?
- Where does a point fall if it has a y-coordinate of zero?
- How can you apply what you know about the quadrants to determine where the points should be plotted?

DIFFERENTIATE INSTRUCTION

Encourage students to label each quadrant before plotting the points to ensure they can accurately determine where each point should go.

5.1.4 GUIDED INSTRUCTION: REFLECTIONS

Component Overview: *Students investigate reflections of ordered pairs on the coordinate plane. First, they examine the locations of ordered pairs that share either an x-coordinate or a y-coordinate. Then students reflect ordered pairs over the x- and y-axis.*

TEACHER MOVES

Notice and Wonder

Engage the class in a Notice and Wonder routine to help students draw conclusions about plotting ordered pairs on the coordinate grid.

- What do you notice about points with the same
- x-coordinate?
- What do you notice about points with the same
- y-coordinate?
- What do you notice about the distance of these points from the axes?

TEACHER MOVES

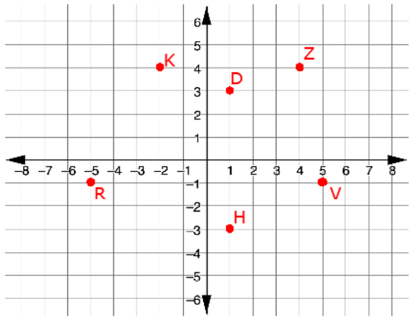
Ask students to discuss strategies to find the reflection of Point T over the y-axis without folding the plane. Once students determine a strategy, prompt them to see if the strategy works for reflections over the x-axis as well.

5.1.4 Guided Instruction: Reflections

1. Use the points in the table for the following.

Point D $(1, 3)$	Point H $(1, {}^{-}3)$	Point K $({}^{-}2, 4)$
Point R $({}^{-}5, {}^{-}1)$	Point V $(5, {}^{-}1)$	Point Z $(4, 4)$

   a. Plot the ordered pairs from the table on the coordinate grid below. Label using the given letter.



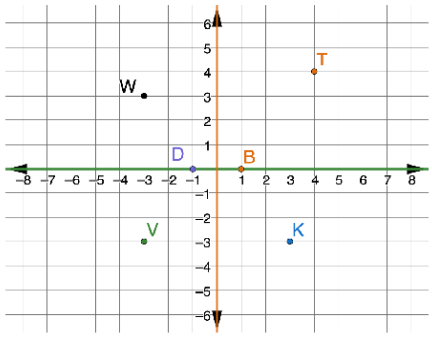
   b. Select the pair of points that have the same x-coordinate.

   ☐D ☐H ☐K ☐R ☐V ☐Z

   c. Select the pair of points that have a y-coordinate of -1.

   ☐D ☐H ☐K ☐R ☐V ☐Z

2. A coordinate grid with points D, B, K, T, V, and W is shown. The x-axis is green and the y-axis is orange.



   a. Write the ordered pair of the reflection of Point V when the graph is folded across the x-axis.
   $(-3, 3)$

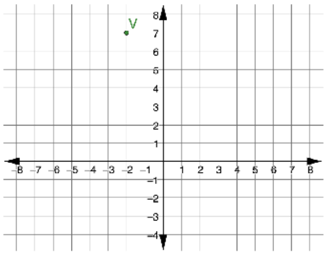
   b. Write the ordered pair of the reflection of Point K when the graph is folded across the y-axis.
   $(-3, -3)$

   c. Write the ordered pair of the reflection of Point D when the graph is folded across the y-axis.
   $(1, 0)$

   d. Write the ordered pair of the reflection of Point B when the graph is folded across the x-axis.
   $(1, 0)$

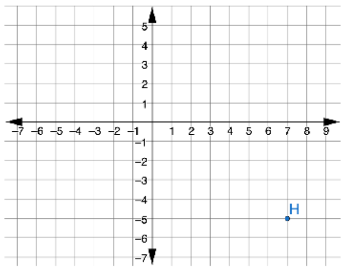
5.1.5 Guided Practice: Reflections

1. Point V is plotted on the coordinate grid.



What are the coordinates of the reflection of V across the y-axis?
$(2, 7)$

2. Point H is plotted on the coordinate grid.



What are the coordinates of the reflection of H across the x-axis?
$(7, 5)$

5.1.5 GUIDED PRACTICE: REFLECTIONS

Component Overview: *Students practice reflecting ordered pairs over the x- and y-axis.*

TEACHER MOVES

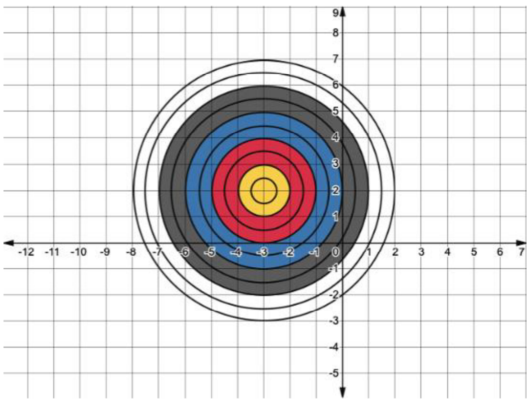
Use this section as an opportunity to clarify misconceptions before moving on to the 5.1.6 Wrap-Up. Use checks for understanding, like asking where either the points or their reflections are located for the coordinates of the plotted points and the quadrants.

QUESTIONING

For an additional challenge, ask students to predict what will happen to the coordinates of an ordered pair after reflecting over both axes.

5.1.6 Wrap-Up: Ticket Out the Door

A target is shown on the coordinate grid.



1. Write the ordered pair that will hit the bullseye.
$(-3, 2)$

2. A player throws two darts. The first dart hits the ordered pair $(-2, -1)$. The second dart hits the point that is the reflection of $(-2, -1)$ over the x-axis. Complete the table that describes the location of the second dart.

Color	Ordered Pair
Red	$(-2, 1)$

5.1.6 WRAP-UP: TICKET OUT THE DOOR

Component Overview: *Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson. This Wrap-Up assesses students' abilities to plot an ordered pair. Students will also be reflecting an ordered pair over the x-axis.*

TEACHER MOVES

Clarify that the bullseye is the very center of a target for students who may be unfamiliar with this vocabulary. In this case, it is the center of the smallest circle in the diagram.